Composite functions

1 Evaluate the following, given the functions:

$$f(x) = 4x^3$$
$$g(x) = x + 5$$

a $f(-2)$ **b** $g(f(-2))$ **c** $f(g(0))$

2 Evaluate $f(g(2))$, given the functions:

$$f(x) = 4x - 10$$
$$g(x) = 3 + \frac{3}{x}$$

3 Consider the following functions:

$$f(x) = -2x - 6$$
$$g(x) = 5x - 7$$

a Find $f(7)$.

b Hence, evaluate $g(f(7))$.

c Find $g(7)$.

d Hence, evaluate $f(g(7))$.

e Does $f(g(x)) = g(f(x))$ for all x?

4 Find the composite function $f(g(x))$ for the following:

a $f(x) = x^3$, $g(x) = 9x - 4$

b $f(x) = x^2$, $g(x) = x^2 + 1$

c $f(x) = \sqrt{x}$, $g(x) = 4x - 3$

d $f(x) = 4x + 9$, $g(x) = x^3$

e $f(x) = 4x - 3$, $g(x) = x^2$

5 Evaluate the following, given the functions:

$$f(x) = -2x + 2$$
$$g(x) = 4x^2 - 8$$
$$r(x) = -3x - 8$$

a $g(6)$ **b** $f(g(6))$ **c** $r(f(g(6)))$

6 Consider the following functions:

$$f(x) = -2x - 8$$
$$g(x) = 4x^2 - 4$$

a Evaluate $g(f(6))$.

b If $h(x)$ is defined as $f(g(x))$, state the equation for $h(x)$.

c What type of functions are $f(g(x))$ and $g(f(x))$?

7 Consider the following functions:

$$f(x) = -2x + 6$$
$$g(x) = 3x + 1$$

a If $r(x)$ is defined as $f\left(x^2\right)$, state the equation for $r(x)$.

b Hence, state the equation for $q(x)$, which is $g\left(f\left(x^2\right)\right)$.

8 Consider the following functions:

$$f(x) = x^2 + 3$$
$$g(x) = 4x - 9$$

a State the equation for $f(2x)$.

b Show that $f(2x) = g(f(x))$.

9 The function $f(x)$ is defined as $f(x) = -3x + 4g(x)$.

a Given that $f(x)$ is a quadratic function, what type of function is $g(x)$?

b If $f(x) = -3x + 20x^2$, find the equation for $g(x)$.

c Find an algebraic expression for the function $g(f(x))$.

10 Consider the following functions:

$$p(x) = x + 3$$
$$q(x) = x^2 - 1$$
$$r(x) = (x + 4)(x + 2)$$

a Rewrite $r(x)$ in expanded form.

b Show that $r(x) = q(p(x))$.

c Write an algebraic expression for $p(q(x))$.

d Find the range of values for which $r(x) < p(q(x))$.

11 Consider the following functions:

$$f(x) = \frac{4}{x}$$
$$g(x) = x^2 + 3$$

Find the following functions:

a $f(x)\,g(x)$ b $f(g(x))$ c $\dfrac{f(x)}{g(x)}$ d $\dfrac{g(x)}{f(x)}$

12 Consider the following functions:

$$f(x) = 9x + 4$$
$$g(x) = x^2 - 2x - 3$$

Find the following functions:

a $f(x)\,g(x)$ b $f(g(x))$ c $\dfrac{f(x)}{g(x)}$ d $\dfrac{g(x)}{f(x)}$

Domain and range

13 For each of the following functions $f(x)$ and $g(x)$:

i Find the composite function $f(g(x))$.
ii State the domain of $f(g(x))$.
iii Find the composite function $g(f(x))$.
iv State the domain of $g(f(x))$.

a $f(x) = \dfrac{9}{1-2x}$
$g(x) = \dfrac{3}{x}$

b $f(x) = \sqrt{x+4}$
$g(x) = x^2 - 4$

14 Consider the function $h(x) = \sqrt{1+\sqrt{1+x}}$ and suppose that $g(x) = \sqrt{1+x}$.

Find $f(x)$ such that $h(x) = f(g(x))$.

15 Consider the function $h(x) = \dfrac{1}{(x-2)^6}$.

a If $g(x) = x - 2$, find $f(x)$ such that $h(x) = f(g(x))$.
b If $f(x) = \dfrac{1}{x}$, find $g(x)$ such that $h(x) = f(g(x))$.

16 Consider the function $f(x) = 5x^2 + 4$. Define $g(x)$ such that $g(f(x)) = f(x)$ for all x.

17 For each of the following functions:

- **i** State the domain and range of $f(x)$.
- **ii** State the domain and range of $g(x)$.
- **iii** Hence, determine the domain and range of $f(g(x))$.

a $f(x) = x^2$
$g(x) = x - 7$

b $f(x) = \sqrt{x}$
$g(x) = x - 3$

c $f(x) = \sqrt{x}$
$g(x) = 4x + 9$

d $f(x) = x^3$
$g(x) = x + 8$

18 Consider the following functions:

$$f(x) = \sqrt{x+5}$$
$$g(x) = 5x - 6$$

- **a** Find an algebraic expression for the function $f(g(x))$.
- **b** Express the domain of $f(g(x))$, using inequalities.
- **c** Find an algebraic expression for the function $g(f(x))$.
- **d** Express the domain of $g(f(x))$, using inequalities.

19 Consider the following functions:

$$f(x) = -2x$$
$$g(x) = \frac{x}{x-5}$$

- **a** Find an algebraic expression for the function $f(g(x))$.
- **b** Determine the value(s) of x that are not in the domain of $f(g(x))$.
- **c** Find an algebraic expression for the function $g(f(x))$.
- **d** Determine the value(s) of x that are not in the domain of $g(f(x))$.

Applications

20 At sale time in a certain clothing store, all dresses are on sale for $\$5$ less than 75% of the original price.

a Write a function $g\left(x\right)$ that finds 75% of x.

b Write a function $f\left(x\right)$ that finds 5 less than x.

c Construct the composite function $f\left(g\left(x\right)\right)$.

d Hence, calculate the sale price of a dress that has an original price of $\$94$.

21 A computer manufacturer sells hard drives to a retail outlet, each at a cost of $\$11$ more than the manufacturing cost. The retail store then sells each hard drive to the public, charging 60% more than they paid to the manufacturer.

a If m represents the manufacturing cost (in dollars), find a function $A\left(m\right)$ which returns the price of buying a hard drive at the retail store.

b Find the price of buying a hard drive at the store, if the manufacturing cost is $\$22.50$.